

22 October 2025

Why do we need multiple layers to go through.

If we use single layer (perceptron) it was unable to resolve complex problem.

Number of layer depends upon of nodes

In real world, there are libraries which help us in coming up with number of layers & neurons. like AutoML or hyperpara

Rule of Thumb (Not always but mostly follow)

Shallow Network

Moderate

Deep $\rightarrow$

1-2 hidden

↓ linear, mild

$\frac{1}{2}$ ① ② $\frac{0}{2}$
Generally even
number of neurons

Moderate 14

3-5 text, image

Deep

2.

10 + layers

text, audio
video

1215 is linearly separable so 2 layers were all right!

if data is complex like many animals passing in

Start with 1 to 2 layers and if accuracy is not good then increase

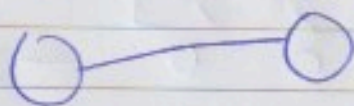
nodes(neurons) 4 layers.

1) Typical number of neurons.

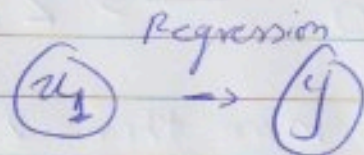
$S_L \rightarrow 0$
 $S_H \rightarrow 0$
 $A_L \rightarrow 0$
 $A_H \rightarrow 0$

~~2³~~ 2^3 and 2^4
8 and 16.

Activation Function (see explanation around 6:38 Am)



$$y = mx + c$$



2¹⁶ values

2¹⁶

$$\frac{1}{e^{-x}}$$

When you're classifying, then see ~~see~~ see

In Classical machine we don't need to compile but for NN we need to compile to build the model.

~~model~~ or other parameters. 6:45
for batch-size we can use libraries
Keras_tuner, ~~optuna~~ - scikit learn. optimiz
optuner

Hyperparameter — number of layers
and others which control
the algorithm.

we used it. (Around 7.06 AM)

$X = d \begin{bmatrix} \end{bmatrix}$ — column format.

$Y = d \begin{bmatrix} \end{bmatrix}$ $(-1, 1)$ will make it column.
↓ row ↓ add 1 dimension

In binary format

$0 \rightarrow \begin{bmatrix} 2^2 & 2^1 & 2^0 \end{bmatrix}$
 0 0 0 $\rightarrow 0$

$1 \rightarrow \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \rightarrow 1$

$2 \rightarrow \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \rightarrow 2$

Softmax — more than 2.

Binary — more than 1

~~Linear~~ — Regression problems.